

Meeting Preparation Notes

Supervisor background & potential project topics, Dr Tianhai Tian, Monash University

Prepared by Jay | 12 August 2026

At a Glance, quick read before you walk in

His current focus isn't just the 5 old topics, he's now also working in deep learning for single-cell genomics (DeepCE, PNAS manifold-distance paper), modern architectures (Kolmogorov-Arnold networks), cancer drug resistance / angiogenesis, and even option pricing. Ask directly what he's currently supervising.

Top things to ask him

1. What is he currently supervising / what's the specific project on offer?
2. Project level, timeline, and funding/scholarship status.
3. What background reading or tools should you start with?

Top things to remember about yourself

1. Your math strengths and honest gaps (see MTH3230 refresher, p.2-3, if he probes).
2. Why this area interests you, have a genuine, specific answer ready (see companion 'about me' template).
3. Any programming/tools experience (MATLAB, Python, R).

Bring / do

Printed CV or transcript if you have one | something to write with | arrive a few minutes early | phone silent.

Supervisor Profile

Dr Tianhai Tian is an Associate Professor and ARC Future Fellow in the School of Mathematical Sciences at Monash University. He completed his PhD in Mathematics at the University of Queensland under Professor Kevin Burrage, and has held research fellowships including the Lord Kelvin Fellowship (University of Glasgow) and an Australian Research Council Future Fellowship. His work sits at the intersection of applied mathematics, statistics, and computational biology.

Position	Associate Professor & ARC Future Fellow
Department	School of Mathematical Sciences, Monash University
PhD	Mathematics, University of Queensland (2001)
Prior roles	Lord Kelvin Fellow (Glasgow); Senior Research Fellow, IMB (UQ)
Email	tianhai.tian@monash.edu

My Prior Coursework with Him, MTH3230

Dr Tian taught MTH3230 (Random Processes for Time Series Analysis) in my first year, second semester. The course covered:

- Probability and random processes, stationary processes
- Forecasting stationary time series (linear prediction, Durbin-Levinson and innovation algorithms)
- ARMA models, causality, invertibility, autocovariance function, partial autocorrelation
- Spectral analysis, spectral densities, nonparametric density estimation, linear operators on time series
- Nonstationary and seasonal time series models, including ARIMA
- State-space models, Wiener filter and Kalman filter

This is a natural bridge into his research: stationary processes, ARMA/spectral methods, and Kalman filtering are all core tools for analysing noisy, time-dependent data, the same kind of thinking behind stochastic modelling of gene expression and signalling dynamics. Worth mentioning this course directly in the meeting as shared ground.

How MTH3230 Connects to Each of His Five Topics

The course gives a general toolkit, probability theory, stationary processes, spectral analysis, and state-space/Kalman filtering, for modelling systems that evolve randomly over time and inferring parameters from noisy data. That underlies his work broadly, but the fit is closer for some topics than others:

Topic	Fit	Connection
5. Telomere length regulation	Strongest	Telomere shortening over cell divisions is essentially a random process observed over time, close to the stationary/nonstationary time series ideas in Ch. 2 and 6.
2. Numerical methods / parameter inference	Strong	His simulated maximum likelihood and Bayesian parameter estimation from time-course data is conceptually close to state-space/Kalman filtering (Ch. 7), inferring hidden states or parameters from noisy, time-indexed observations.
3. Cell signalling pathways	Moderate	Models are fitted to proteomic/microarray time-course data; spectral density estimation and forecasting ideas could apply to characterising that experimental data, though it's not his primary tool.
1. Gene regulatory networks	Indirect	Relies mainly on stochastic simulation (Gillespie SSA, chemical master equations) rather than ARMA/spectral methods, shared probability theory foundation, not shared specific techniques.
4. Cancer robustness / therapy	Indirect	Leans on stochastic models, dynamical systems and PDEs, again shared foundations in random processes rather than direct method overlap.

Useful to be precise about this in conversation rather than claiming a blanket connection, it shows genuine understanding rather than a surface-level link.

Quick Refresher, Basic MTH3230 Q&A

In case he checks your grasp of the fundamentals from his course:

Q: What is a stochastic process?

A: A sequence of random variables indexed by time, e.g. $X_1, X_2, \dots$. Before you measure anything it's random; once observed, you get one realised sequence of numbers called a sample.

Q: What does it mean for a process to be stationary?

A: In the wide/weak sense used in this course: the mean is constant over time, and the covariance between X_n and X_m depends only on the time gap ($n - m$), not on where you are in time. This lets you use one dataset to estimate properties that are assumed to hold at every point in time.

Q: What is white noise?

A: A sequence of zero-mean random variables that are uncorrelated with each other (orthogonal). It's the basic 'building block', ARMA models are constructed by passing white noise through linear filters.

Q: What are the autocovariance (ACVF) and autocorrelation (ACF) functions?

A: The ACVF, $\gamma(h)$, measures the covariance between observations h steps apart; the ACF, $\rho(h) = \gamma(h)/\gamma(0)$, is the same thing normalised to lie between -1 and 1. They summarise how strongly a series is correlated with its own past.

Q: What's the difference between AR, MA, and ARMA models?

A: An AR(p) process expresses X_n as a linear combination of its own past p values plus noise. An MA(q) process expresses X_n as a linear combination of the past q noise terms. ARMA(p, q) combines both. ARMA models are flexible enough to approximate almost any stationary time series.

Q: What do 'causal' and 'invertible' mean for an ARMA process?

A: Causal means X_n can be written using only present and past noise terms (not future ones), required for the process to be realistically forecastable. Invertible means the noise term Z_n can be recovered from present and past X values. Both properties come down to where the roots of the model's characteristic polynomials lie relative to the unit circle.

Q: What is the partial autocorrelation function (PACF) and why use it?

A: The ACF can show a false correlation between X_n and X_{n-2} purely because both are linked through X_{n-1} . The PACF strips out that indirect chain of dependence, isolating the direct relationship, useful for identifying the order p of an AR model.

Q: What if a series isn't stationary, what do you do?

A: Difference it. Working with $Y_n = X_n - X_{n-1}$ (or higher-order differences) often makes an otherwise trending/non-stationary series stationary, after which an ARMA model is fitted to the differenced series, this is the idea behind ARIMA(p, d, q) models.

Q: What is spectral density, in plain terms?

A: It re-expresses the autocovariance structure of a stationary process in the frequency domain, essentially decomposing the series into how much variation happens at each frequency/cycle length, rather than looking at correlations lag by lag in time.

Q: What is a state-space model / Kalman filter used for?

A: A framework for tracking a hidden (unobserved) state that evolves over time, using noisy observations of it. The Kalman filter recursively updates the best estimate of that hidden state as new data arrives, directly relevant to inferring unobserved biological quantities from noisy experimental time-course data.

Q: What does 'causal' mean precisely, in terms of the model's polynomial roots?

A: For an ARMA process built from the autoregressive polynomial $\Phi(z)$, causality holds if all roots of $\Phi(z)$ lie outside the unit circle ($|z| > 1$). This guarantees X_n can be expressed using only present and past noise terms, so the model is forecastable from history alone.

Q: What does 'invertible' mean precisely?

A: Invertibility is the mirror condition on the moving-average polynomial $\Theta(z)$: all its roots must lie outside the unit circle. This ensures the noise term Z_n can be reconstructed from present and past values of X , which matters for parameter estimation and forecasting.

Q: What is the projection theorem / best linear predictor, and why does it matter?

A: Forecasting is framed as finding the point in a subspace (spanned by past observations) closest to the true future value, in a mean-squared-error sense. The projection theorem guarantees this best predictor exists and is characterised by the forecast error being orthogonal (uncorrelated) to every observation used to build it, this is the basis for the Durbin-Levinson and innovations algorithms.

Q: What's the difference between the population ACVF and the sample ACVF?

A: The population ACVF, $\gamma(h)$, is a theoretical property of the model. The sample ACVF, an average of $(x_n - \bar{x})(x_{n+h} - \bar{x})$ over the observed data, is what you actually compute from a dataset to estimate $\gamma(h)$, accuracy improves as the sample size grows.

Q: How do you handle a series that has both a trend and a seasonal pattern?

A: Decompose it as $X_n = \text{trend} + \text{seasonal} + \text{stationary residual}$, remove the trend and seasonal components first (e.g. via regression or differencing), then fit an ARMA model to the leftover stationary residual.

Q: What's a seasonal difference and when is it needed?

A: An ordinary difference subtracts the previous observation ($X_n - X_{n-1}$); a seasonal difference subtracts the observation one full period earlier ($X_n - X_{n-s}$), e.g. $s = 12$ for monthly data with yearly seasonality. It's used when regular differencing alone doesn't remove periodic structure, as with data like milk production.

Q: What is the periodogram used for?

A: It's the standard nonparametric estimate of the spectral density, computed directly from the data (via something equivalent to a discrete Fourier transform of the sample), used to see which frequencies dominate a time series without assuming a specific ARMA model first.

Q: Why model data with ARMA instead of just describing it directly?

A: A fitted model compresses the data into a small number of interpretable parameters, allows principled forecasting with quantifiable uncertainty, and lets you separate genuine signal from noise, raw descriptive statistics alone can't forecast or generalise beyond the observed sample.

Important Update, His Current Research (2024-2025)

His personal homepage is outdated (last updated ~2015). His official Monash research profile shows his focus has broadened well beyond the five topics listed there, now includes deep learning and single-cell genomics alongside the original stochastic/cancer modelling work. Recent outputs (2024-2025):

- **Diffusive topology preserving manifold distances for single-cell data analysis**, PNAS, Jan 2025 (single-cell RNA-seq analysis method).
- **DeepCE: a deep learning framework for correlation-enhanced gene regulatory network inference in single-cell RNA sequencing data**, Bioinformatics Advances, 2025/2026 (deep learning for gene network inference).
- **Kolmogorov-Arnold network-based enhanced fusion transformer for hyperspectral image classification**, Information Sciences, Nov 2025 (modern deep learning architectures).
- **Mathematical modeling of drug resistance in heterogeneous cancer cell populations**, International Journal of Biomathematics, Feb 2025.
- **Dynamic modeling and analysis of tumor angiogenesis and anti-angiogenesis treatment**, Journal of Biological Systems, Jun 2025.
- **Adaptive option pricing based on a posteriori error estimates for fully discrete finite difference methods**, Journal of Computational and Applied Mathematics, May 2025 (mathematical finance).

Takeaway: don't treat the five listed topics as the full picture. He's actively working with deep learning (transformers, Kolmogorov-Arnold networks) applied to single-cell genomics and image classification, plus cancer modelling (drug resistance, angiogenesis) and even mathematical finance. Worth asking directly what he's currently supervising rather than assuming it's one of the five.

Core Research Interests

- Stochastic modelling of genetic regulatory networks, how randomness in gene expression produces switch-like and oscillatory behaviour in cells.
- Mathematical modelling of cell signalling pathways, particularly the MAP kinase cascade and Ras-driven signal transduction.
- Cancer robustness and therapy resistance, why tumours survive treatment and how resistance evolves.
- Telomere length regulation in ageing, mathematical models of telomere shortening and cell fate.
- Numerical and computational methods, efficient stochastic simulation algorithms (SSA, tau-leap, delay-SSA) for large biological systems, and parameter inference from omics data.

His Five Postgraduate Project Topics

Listed on his research projects page as candidate PhD/Master's topics for prospective students:

1. Stochastic Modelling of Genetic Regulatory Networks

Gene expression is inherently noisy, driven by small molecule counts and random biochemical reactions. This project develops stochastic (rather than purely deterministic) models to explain

experimentally observed behaviours such as bistability and oscillation, applied to systems like the lactose operon (*E. coli*) and the p53 tumour-suppressor network.

2. Effective Numerical Methods for Simulating Chemical Reaction Systems

Exact stochastic simulation (the Gillespie SSA) is accurate but computationally expensive at scale. This project focuses on designing faster simulation algorithms, such as binomial tau-leap and multi-scale methods, enabling large, realistic biological networks to be simulated efficiently. A strong fit for candidates with a computational/numerical-methods background.

3. Mathematical Modelling of Cell Signalling Pathways

Cells transmit external signals (e.g. growth factors) into internal responses via cascades of interacting proteins, most notably the MAP kinase pathway. This project builds models, often fitted to real proteomic/microarray data, to understand signal propagation and cross-talk between pathways, with direct relevance to cancer biology.

4. Mathematical Modelling of Cancer Robustness and Cancer Therapy

Tumours are remarkably resilient to aggressive treatment, and resistance can be present from the outset or evolve during therapy. This project applies stochastic models, dynamical systems, and PDEs to understand the mechanisms behind tumour robustness and to inform better-targeted therapies.

5. Mathematical Modelling of Telomere Length Regulation in Ageing Research

Telomeres shorten with each cell division and are linked to cellular ageing and disease. This project develops mathematical models of the regulatory mechanisms governing telomere length and its role in determining cell fate, a comparatively newer, less-developed strand of his research.

Note: his publication list ends around 2014-2015 and the listed grants have mostly concluded, so his current focus may have shifted since this page was last updated, worth asking directly.

Logistics to Clarify in the Meeting

- Project level and mode, Honours, Master's, or PhD; full-time or part-time.
- Timeline and milestones, expected duration and key checkpoints.
- Funding / scholarship status, if relevant.
- Meeting cadence, how often, and in what format (in person / online).
- Co-supervisors, will anyone else be involved in supervision?
- Data / resources, is there an existing dataset or system already identified, or is that still open?
- Tools required, software or programming background needed (e.g. MATLAB, R, Python, SSA implementations).
- What a successful outcome looks like by the end of the project.

Notable Publications to Reference

A few of his papers worth naming naturally in conversation (not reciting a list), early landmark work plus his most recent output:

Landmark earlier work

- Tian T. & Burrage K., *Stochastic models for regulatory networks of the genetic toggle switch*, PNAS, 103, 8372-8377, 2006, a foundational paper in stochastic gene-network modelling.
- Tian T., Harding A., Inder K., Plowman S., Parton R.G. & Hancock J.F., *Plasma membrane nano-clusters generate high-fidelity Ras signal transduction*, Nature Cell Biology, 9, 905-914, 2007.
- Tian T., Plowman S., Parton R.G., Kloog Y. & Hancock J.F., *Mathematical modelling of K-Ras nanocluster formation on the plasma membrane*, Biophysical Journal, 99(2), 534-543, 2010.
- Wang J. & Tian T., *Quantitative model for inferring dynamic regulation of the tumour suppressor gene p53*, BMC Bioinformatics, 11(1), 36, 2010.
- Tian T., Olson S., Whitacre J.M. & Harding A., *The origins of cancer robustness and evolvability*, Integrative Biology, 3(1), 17-30, 2011.
- Tian T., *Chemical memory reactions induced bursting dynamics in gene expression*, PLoS One, 8(1): e52029, 2013.

Most recent work (2024-2025)

- Wei J., Zhang B., Wang Q., Zhou T., Tian T. & Chen L., *Diffusive topology preserving manifold distances for single-cell data analysis*, PNAS, 122(4), e2404860121, Jan 2025.
- Wu Q., Dai X., Lou S., Wu S. & Tian T., *DeepCE: a deep learning framework for correlation-enhanced gene regulatory network inference in single-cell RNA sequencing data*, Bioinformatics Advances, 6(1), vbag033, 2025/2026.
- Bao K., Liang G., Tian T. & Zhang X., *Mathematical modeling of drug resistance in heterogeneous cancer cell populations*, International Journal of Biomathematics, 18(2), 2350101, Feb 2025.
- Pang L., Liu S., Liu C., Hu S., Zhang Z. & Tian T., *Dynamic modeling and analysis of tumor angiogenesis and anti-angiogenesis treatment*, Journal of Biological Systems, 33(2), 647-667, Jun 2025.
- Han X., Jiang F., Wen S. & Tian T., *Kolmogorov-Arnold network-based enhanced fusion transformer for hyperspectral image classification*, Information Sciences, 717, 122323, Nov 2025.
- Mao M., Wang W., Tian T. & Wang L., *Adaptive option pricing based on a posteriori error estimates for fully discrete finite difference methods*, Journal of Computational and Applied Mathematics, 460, 116407, May 2025.

Tip: don't just name-drop titles, pick one or two that connect to a topic you're genuinely curious about and ask a follow-up question. E.g. 'I saw your PNAS paper on single-cell manifold distances, is that connected to any of your gene network projects?' lands far better than reciting the list.

Sources: users.monash.edu.au/~tianhai/ ; research.monash.edu/en/persons/tianhai-tian

Your “Tell Me About Yourself” Answer, Ready to Practice

Based on what you told me. Don't recite this word-for-word, read it a couple of times out loud, then let it come out naturally in your own words.

1. Background

“I'm doing a Master's in Mathematics. In my Bachelor's I studied some basic statistics, and I've actually been teaching statistics at school level and coaching Bachelor's students in statistical subjects, so I've spent a fair bit of time explaining these ideas to others, not just learning them myself. I took your MTH3230 unit, Random Processes for Time Series Analysis, and that's genuinely where things clicked for me. I was already interested in statistics from my Bachelor's, but that course sharpened it into a specific interest in random processes and time-dependent data.”

2. Strengths and honest gap

“Statistics is my strongest area, especially after MTH3230, stationary processes, ARMA models, that whole way of thinking about noisy, time-evolving data. The one gap I have is biology, I don't have any formal background there. But I'm genuinely keen to pick it up if it's needed for the project; I'd rather be upfront about that than pretend otherwise.”

3. Why this area / why him

“Looking at your project list, two topics stood out to me, the numerical methods for simulating stochastic reaction systems, and the telomere length regulation work. Both feel like a natural extension of what I learned in your course: the telomere project is essentially a random process observed over time, which is very close to the stationary/nonstationary time series ideas from MTH3230, and the numerical methods project is more computational and math-heavy, which suits where my strengths currently are given I don't have a biology background yet.”

4. A question of your own

You said you don't have one yet, that's completely fine, better to ask a real question in the room than force one now. If you want a safe, genuine one ready as a fallback, given your own gap:

“For a project like telomere modelling or the numerical methods work, how much biology background would I need going in, is that something I'd pick up along the way, or should I start reading into it now?”

This is a good fallback because it's honest (ties to your actual gap), shows you're already thinking practically about the project, and isn't a fabricated “insight” question, it's one you actually need answered.

Reminder: it's fine to explicitly mention Project 2 and Project 5 by name in the meeting, you have a genuine reason for both (see the connection table earlier in this document).